LDA: Jordan (Good, 1950s)

Multinomial likelihood.

$$\frac{1}{\Gamma(\cdots + \alpha_1 + \cdots \alpha_K)} \theta_1^{\alpha_1 - 1} \cdots \theta_K^{\alpha_K - 1}$$

learn α's.

Topic Models / NLP

Larry Carin (duke)

Beta, Gamma, ...
Dirichlet.

$$\frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}$$

learn α.

Hartman (76).

$$\frac{e^{-b\alpha}}{\Gamma(1+\alpha)} = \int_0^\infty e^{-w\alpha^2} p(w)\, dw$$

completely monotone.

latent variable data augmentation.

ER-Gamma (Polya-Gamma)

$$e^{b\theta}/(1+e^\theta)^c$$

LPU

MCMC: $p(w|\alpha) = \dfrac{e^{-w\alpha^2} p(w)}{e^{-b\alpha}/\Gamma(1+\alpha)} \sim w^{[g]}$?

indep.

saddlepoint

Windle, ... (2014)

EM: data augmentation (Polson-Scott, Brometulea)

$\rightarrow E(w|\ldots) = w^{[g]} \longrightarrow$ digamma Γ', Minka Thomas.

Inverse Dirichlet

ER-Gamma!

$\beta | w \rightarrow$ conditionally normal

$\rightarrow$ SGD updates in blocks.

$w | \beta \rightarrow$

- Data Augmentation
- ADMM
- PGamma.

E-RGamma

$y^{-s-3/2}$

$$\frac{c\, e^{-a\theta}}{\Gamma(1+\theta)}$$

Exp-RGamma dist.

$a < \gamma$

choice

NCT

Beta-Bin-Hollingsworth

LDA

SMN.

$$\int_0^\infty e^{-w\theta^2} p(w)\, dw \qquad (2*)$$

$p(\lambda | y)$

$w \sim ERG(a, 0)$

$$p_{a,b}(\theta) = c_{a,b} \frac{e^{-a\theta}}{\Gamma(b+\theta)}$$

$EG(a,b)$.

$$= c_{a,b} \int_0^\infty e^{-w\theta^2} p_{a,b}(w)\, dw$$

$$p(w|\theta) = \frac{e^{-w\theta^2} p_{a,b}(w)}{E(\text{—})} \sim w^{(g)}$$

Robert

Thomas Minka

EM

polson-scott data-augmentation

Robert Lloyd George (grandson David Lloyd George)

New China Era fund (holdings) (investing)

Jan 1, 18. Wuxi!

Bamboo fund, India — ? (changes) in holdings

---

bilibili (holders).

Whalewisdom (hedge funds).

---

shenzen - Hys —?

guangdong -

Hatteewizards (Mew) — Schwager (1991).

Bill Ackman (300)

Stan Druckenmiller.

Arise Learning!

T-D-T

Economic of Nutlicker

Tuwou. 1950s.

Likelihoods. Beta, Gamma, Topic Model, LDA, Dirichlet, NP-Bayes

(VB)

LSTM, JBES

$\frac{1}{\Gamma(a+y)}$

lean a.

Weir & Escobar (1995) Beta latent model.

Minka Bayesian Dirichlet: MCMC EM for "a".

scalable. Hartman 1976 seeing differential equ.

not closed form

MCMC: $\frac{e^{-\gamma a}}{\Gamma(a+y)} = \int_0^\infty e^{-a^2\lambda} P_y(\lambda)\, d\lambda$. Devroye

XB

scale mixture

mgf

= exponential gamma SGD ⇒ Gibbs. Windle,

Exp-Reciprocal Gamma (ERG) — $\lambda | a \sim GGC$ Lennart Bondesson (1992).

Generalized Gamma Convolution (GGC)

→ Lévy measure → limits of compound Poisson

$E[e^{-s\lambda}] = \exp\left(\int_0^\infty (1-e^{-st}) \frac{\mu(dt)}{t}\right)$

Ferguson & Klass: —

$\times \times \times \times \times \times$ J

NO

X.

$\sum_j e^{-}$

Weinning Shen Ime

$\frac{1}{\Gamma(1+\sqrt{\mu})} = E[e^{-\mu\lambda}]$ Bernstein fn

CM

$$p(a|y) = \frac{1}{\Gamma(a+y)} \cdot p(a)$$

sample

$$\frac{e^{\mu b}}{\Gamma(1+\mu)}$$

b

K

$$p(a,\lambda|y) = \left\{ e^{-a^2\lambda}\, p(\lambda) \right\} p(a)$$

latent variable

$$p(a,\lambda|y) = e^{-a^2\lambda} p(\lambda)\, p(a)$$

N

$$p(\lambda|a,y) \sim e^{-a^2\lambda} p(\lambda)$$

$$p(a|\lambda,y) \sim \text{Normal}$$

Ferguson+Klass

Exp. Tilt.

$$\lambda \overset{D}{=} \sum_{n=1}^{\infty} \frac{\varepsilon_n}{q_n}$$

GGC.

limit Poisson

mgf.

$$E_{\lambda|a,y}\{e^{\lambda s}\} = M(s)$$

mgf

$$f(\mu) = \frac{t^{\mu}}{\Gamma(1+\mu)} = \exp(-\rho(\mu))\,(te^{\gamma})^{\mu}$$

$$= \exp \int_0^{\infty} (1-e^{-})\,\mu(-)\,d-$$

$$\varphi(\mu) = \rho(\sqrt{\mu}), \quad \frac{d\varphi}{d\mu} = \sum_{n=0}^{\infty} \frac{1}{2(n+1)(n+1+\sqrt{\mu})}$$

$$\frac{e^{-\mu^2 c}\, t^{\mu}}{\Gamma(1+\mu)} = \exp\left\{ \mu \int_0^\infty (1-e^{-s\mu})\, \nu(s)\, ds \right\}$$

$c\mu^2 +$ Lévy measure

Dennye.

$$\exp\left\{ -c\mu^2 - b\mu + \int_0^\infty (1-e^{-s\mu})\, \nu(s)\, ds \right\}$$

$b \to 0.56$

$\int_0^\infty \min(1,s)\, \nu < \infty$

Bionehler. Walker-Dene.

Mattia Winkel oxford Lévy

$$p(\mu|b) = \frac{C\, e^{-\frac{a}{2}\mu^2}\, e^{-b\mu}}{\Gamma(1+\mu)}$$

γ

Bamboo Asia Food

China News

Robert Lloyd George

$$\approx \exp\left\{ -\mu^2 \text{———} \right\}$$

0.56

Fanhua

$$\frac{e^{-\frac{1}{2}\mu^2}}{\Gamma(1+\mu)} = \frac{e^{-\gamma\mu}}{\Gamma(1+\mu)}\, e^{\gamma\mu - \frac{1}{2}\mu^2}$$

Binet.

$$\Gamma(1+\mu) = \exp - \int_0^\infty (1-e^{-s\mu}) \frac{e^{-s}}{s} \, ds$$

Łeuy nerone

$$\frac{d}{d\mu} \nearrow \psi(1+\mu)$$

$$\log \frac{1}{\Gamma(1+\mu)} = \int_0^\infty (1-e^{+s\mu}) \frac{e^{-s}}{s} \, ds$$

$$\frac{d}{d\mu} \quad \varphi(1+\mu) = \text{———}$$

Bayesian Analysis of Exponential Reciprocal Gamma Models (ERG)

Simulation: Bondesson (1982). Shot-noise process representation of ID r.v.s

GGC can be approximated by a compound Poisson Process.

GGC where Lévy measure has Thorin $U(dt)$ representation.

see Section 4.3. Bondesson.

sequential learning

Data Augmentation

James (2007). Gamma Process and approx. GGC simulation.

Representation: Roynette & Yor (2005). $\frac{e^{cs}}{\Gamma(1+s)}$, $c \le -\gamma$ $= E[e^{-s^2 H}]$

Lemma 1 (Polson, 2018). Finds $U(dt)$ from $\mu(dt)$ ↳ Lévy measure.

Need: $\frac{1}{\Gamma(a+s)}$ → "learning" s. Normal Prior on "s".

→ "data" Simulation with extra normal component

see Roynette & Yor.

→ convert this to a simulation algorithm (Bondesson),

Models. List of Bayes/Machine learning Models which use reciprocal Gamma functions. Pitman-Yor / Genetics ... Spectral Models.

or seq. learning

Dirichlet / Multinomial / Topic Models. Long list. NP-Bayes

LDA

West & Escobar (1995) JASA.

Ferguson & Klass (1972?) Shot noise. ecommerce. "α"

References

Bondesson (1982) (1992, book)

~~Roynette & Yor (2005)~~ – Damien, Bionetrika – JRSSB

Hartman (1976)

Polson, Scott, Windle (2012) Pólya-Gamma

Windle, Polson, Scott (2014) Lijoi

West & Escobar (1995)

James (2007). GGC & Dirichlet.

Minka (2003?) Thomas. Polson & Scott (2012) (SYM) BA

EM, ECM, ECME.

$c \le -\gamma$

$$\frac{e^{cs}}{\Gamma(1+s)} = E[e^{-s^2 H}] + \text{Prior}$$

data

Hartman, Roynette & Yor (2005)

mgf

$$\exp\left\{ \int_0^\infty (1-e^{-st}) \frac{\mu(dt)}{t} \right\}$$

Lévy

ID $\quad (\tau s^2)$ $\quad E_U[e^{-tU}]$

G.G.C. $\quad$ Thorin.

$$\mu(t) = \int_0^\infty e^{-ut}\, U(du)$$

1. Introduction 1.1 Connection with Machine Learning List / Table Models PG.

2. Exp. Reciprocal Gamma (ERG) 2.3. Data Augmentation EM / MCMC.
2.1. Roynette, Yor, Hartman
2.2 Posterior Dist (Bayes learning)

3. Simulation Lancelot Add normal prior
Boudesson, James, Ferguson & Klass, ... Damien, Smith, Walker

4. Application
Dirichlet / Multinomial (LDA). Jack Good, Lindley.

5. Discrim.
Data Augmentation for Gammatus. Beta-Neg Bin (Gamma-) Mingyuan Zhou.
Augmentation Scheme for particle MCMC

Data: BikeSharing data in R

Didonis & Chen. (2009, JASA).
Contingency tables.
Beta Prior → Gamma.
Natesh Pillai. S. Walker.
Robert Stan.

Jasa
JRSS
Doss: Bayesian Poisson Regression
Kim et al, JASA, 2013.

P. Dellaportas (gamma) seq. learning → PFs. Biomedicine
P. Fearnhead. model choice. Fearnhead & Clifford.

D. Damien. (gamma — ....)
H. Doss

JRSS '61
log-well
Poisson reg.
Chopin.

Global-local shrinkage Prior → Scale Mix Normal
Likelihood = Scale mixture

Horseshoe + Topic (you —)

KF.

α_1
α_L

$p_1^{-1} \cdots p^{-1}$ Chamberlain

Chamberlain

Hirano, Book. shrink α

1. Literature Review

List Models .... NP Bayes
LDA
Beta Priors/Gamma —
⋮

2. Simulator

Explain Bondesson. 1-d example.
Posterior. (grid comparison)

↓

Ferguson + Klass
⋮

3. Bike-share Time Series Count.
Bayesian Poisson regression.

(or α's → shrinkage) + Global-local Priors etc —
(or particle filter + FFBS, conditional KF)
Lopes.

Topic

Connections

Like PG & SVM, ...
Like Munka, EM.
Exact sampling GLM.
~ of Gamma!

↓

Not VB.

Dir-Multinomial

(1

$$p(X|\alpha) = \frac{\Gamma(\sum_k \alpha_k)}{\Gamma(N+\sum_k \alpha_k)} \cdot \prod_k \frac{\Gamma(N_k+\alpha_k)}{\Gamma(\alpha_k)}$$

$$= p(\alpha)$$

$$e^{-\gamma \sum_k \alpha_k}$$

$\gamma = \psi'(1)$

Euler

≈ 0.57.

$$p(\alpha|X) = \frac{\Gamma(\sum_k \alpha_k)\, e^{-\gamma \sum \alpha_k}}{\Gamma(N+\sum_k \alpha_k)} \cdot \prod_k \frac{\Gamma(N_k+\alpha_k)}{\Gamma(\alpha_k)}$$

ERG-distr $\sum \alpha_k$.

Roynette & Yor

$$E\left[\gamma_{\alpha_k}^{N_k}\right]$$

Minka (2003).

$$= E\left\{ \exp -\left(\sum_k \alpha_k\right)^2 H \right\} \qquad E\left[\gamma_{\alpha_k}^{N_k}\right]$$

$\hookrightarrow$ $\alpha_k | H \sim$ Truncated Normal $\alpha_k \geq 0$.

"$\frac{\Gamma(\alpha)}{\Gamma(\alpha+s)} e^{-\gamma s} = E\left[e^{-s^2 H_k}\right]$"

$\alpha = \sum_k \alpha_k$.

Roynette & Yor

Waldcopler.

$$\frac{e^{-\gamma \sum \alpha_k}}{\Gamma(N+\sum \alpha_k)} \cdot \prod_k \frac{\Gamma(N_k+\alpha_k)}{\Gamma(\alpha_k)} \cdot \Gamma\left(\sum_k \alpha_k\right)$$

$$\| $$

$$E\left[\gamma_{\alpha_k}^{N_k}\right]$$

$$\int_0^\infty u^{N_k} \cdot u^{\alpha_k - 1} e^{-u}\, du$$

independent variables

$$\Gamma(\alpha) = \int_0^\infty x^{\alpha-1} e^{-x}\, dx$$

$$\Gamma(\alpha) = E\left[\gamma^{\alpha}\right].$$

$(\alpha_1, \ldots, \alpha_k | u_1, \ldots, u_{k+1}, H)$

$(u_1, \ldots, u_k | \alpha_1, \ldots, \alpha_k)$

$(u_1, \ldots, u_{k+1}, H)$ ERG

$H | \sum_{k=1}^{K} \alpha_k$

Bondon

$$\frac{e^{-\gamma \sum \alpha_k}}{\Gamma(N+\sum \alpha_k)} \cdot \prod_{k=1} \frac{\Gamma(N_k+\alpha_k)}{\Gamma(\alpha_k)} \cdot \Gamma\left(\sum_k \alpha_k\right)$$

$\sum \alpha_k \le -$

$$e^{-H(\sum \alpha_k)^2} \qquad \prod_k u_k^{N_k+\alpha_k-1} \cdot e^{-u_k} \cdot u_{k+1}^{\sum \alpha_k - 1} e^{-u_{k+1}}$$

$p(u_1, \dots u_{k+1}, H, \alpha_1 - \alpha_k | x)$.

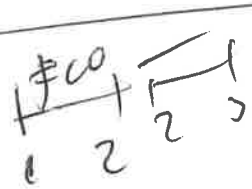
$e^{\alpha_k \log u_k}$

$\alpha_k \ge 0$

Robert

$\exp\{ Q(\alpha_1 - \alpha_k) \}$

Chopin



$\alpha_1 - \alpha_k \sim$ Normal.

$\alpha_k > 0$.

Richard Davis

Gibbs

$\alpha_1 - \alpha_k$

$e^{-H(\sum \alpha_k)^2} p(H)$

exp. tilt.

$\Rightarrow$ Bondesson

$E[e^{tH}] = \exp(\quad)$

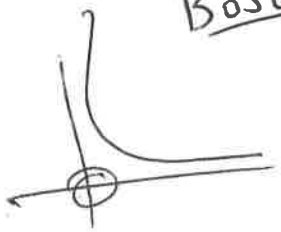
Lévy

= Shot noise.

compound Poisson.

Emily Fox, Mike West, Teh.

NP-Bayes, NLP!



Boscardin, Gelman (2006)

Gibbs

par. Tra

Bondesson

$$E(e^{-sX}) = \varphi(s) = \exp\left\{-as + \int_0^\infty (e^{-sy}-1)\,N(dy)\right\}$$

Lévy measure

ID dist

if $E[e^{-yU}]$ Thorin/GGC

mixture exponentials

$g(u)$

Shot noise

$$X = \sum_{i=1}^{\infty} Z(T_i) = X(0) \text{ where } X(t) = \sum_{T_i \le t} Z(t-T_i)$$

T_i points $P_0(\lambda)$-process.

Truncate: $X_T = \sum_{T_i \le T} Z(T_i)$, $\quad E(e^{-sX_T}) = \exp\left\{\lambda \int_0^T (S(s,u)-1)\,du\right\}$

$$E[\exp(-sZ(u))]$$

compound Poisson

Want to simulate X

$$= \exp\left\{\lambda \int_0^\infty (e^{-sy}-1) \int_0^\infty H(dy;u)\,du\right\}$$

We match: $\lambda \int_0^\infty H(dy;u)\,du = N(dy) = E[e^{-dy\cdot U}]$

$g(u)$ $\quad \frac{e^{\gamma s}}{\Gamma(1+s)}$

$H(y/g(u))$ $\quad g \ge 0$.

Then $Z(u) = g(u)V(u)$ where $V(u)$ compts iid with H

Pick $\bar{H}(x) = e^{-x}$ and find g s.t. $\lambda \int_0^\infty e^{-x/u} g^{-1}(du) = \bar{N}(x)$.

Thorin (Section 4.3 of Bondesson pape).

Ferguson-Klass special case.

$$\sum_{i=1}^{T} Z(T_i) \stackrel{D}{=} X.$$

Tom Ferguson, (UCLA)

Chris Gane

Bondesson

$$E(e^{-sX}) = \varphi(s) = \exp\left\{-as + \int_0^\infty (e^{-sy}-1)\, N(dy)\right\}$$ ID dist

$$X = \sum_{i=1}^{\infty} Z(T_i) = X(0) \quad \text{when} \quad X(t) = \sum_{T_i \le t} Z(t - T_i)$$

T_i : Poisson (λ) process.

Truncate $X_T = \sum_{T_i \le T} Z(T_i)$. $E(e^{-sX_T}) = \exp\left\{\lambda \int_0^T (\varphi(s,u) - 1)\, du\right\}$

$E(\exp(-sZ(u)))$

Simulate X

$$= \exp\left\{\lambda \int_0^\infty (e^{-sx} - 1) \int_0^\infty H(dy, u)\, du\right\}$$

Match.

$$\lambda \int_0^\infty H(dy, u)\, du = N(dy)$$

$$\downarrow$$

$$H(y \mid g(u)) \qquad g \ge 0.$$

Then $Z(u) = g(u) V(u)$. $V(u)$ compts iid with H

Pick $\hat{H}(x) = e^{-x}$ and find g s.t. $\lambda \int_0^\infty e^{-\frac{x}{g}} g^{-1}(dv) = \hat{H}(x)$

Ferguson's special case.

Polya Distribution

Minka.

Dichlet. Minka 2003.

$$P(\underline{x} \mid \underline{p}) = \prod_{k=1}^{K} p_k^{N_k} \qquad N_k = \sum_i \delta\{x_i = k\}.$$

$$P(N_1 \cdots N_K \mid \underline{p}) = \binom{N}{N_1 \cdots N_K} \prod_{k=1}^{N} p_k^{N_k}$$

$$P(\underline{p} \mid \alpha) \sim D(\alpha_1 \cdots \alpha_K) = \frac{\Gamma(\sum_k \alpha_k)}{\prod_k \Gamma(\alpha_k)} \prod_k p_k^{\alpha_k - 1}$$

Posterior

$$P(p \mid x, \alpha) \sim Dir(N_k + \alpha_k)$$

$$P(\underline{x} \mid \alpha) = \underbrace{\frac{\Gamma(\sum_k \alpha_k)}{\Gamma(N + \sum_k \alpha_k)}}_{\text{ERG}} \prod_k \underbrace{\frac{\Gamma(N_k + \alpha_k)}{\Gamma(\alpha_k)}}_{\text{Raynette Yor.}} \cdot \underbrace{e^{-\gamma \sum_k \alpha_k}}_{p(\alpha)}$$

Scale mixture normals

GGC/Lévy measure. Bondesson.

$$E[e^{-sX}] = \varphi(s) = \exp\left\{-as + \int_0^\infty (e^{-sy}-1)\,N(dy)\right\}$$

$E[e^{-yU}]$.

IF: $H(\cdot)$ s.t. $\lambda \int_0^\infty H(dy;u)\,du = N(dy)$

$H(dy;u) = H(y/g(u))$

↳ scale. Given N find g.

or if $H(x) = e^{-x}$

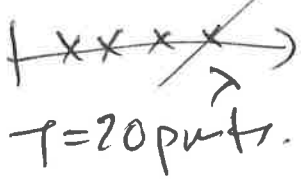
Thorin.

so $\lambda \int_0^\infty e^{-x/u}\, g^{-1}(du) = N(x)$

$\lambda = 1$.

Simulate: $X_T = \sum_{T_i \le T} Z(T_i) + \text{approx(Normal(}$ $\to g(u)V(u)$. skip unless.

$Poi(\lambda)$.



$T = 20$ pts.

Thorin.. $\exp - \int_0^\infty \log\left(1 + \frac{s}{\tau}\right) U(d\tau)$

$= g^{-1}(du)$

finds U.

Hartman.

$$\frac{1}{\Gamma(a+s)} = \exp \int_0^\infty (e^{-sx} + \lambda x - 1)\, \frac{e^{-ax}}{x}\, dx \qquad \text{Lévy}$$

digamma

Malmstem.

digamma

$$\int_0^\infty (1 - e^{-\frac{1}{2}s^2 t}) \cdot \frac{1 - e^{-\frac{1}{2}\frac{x^2}{t}}}{\sqrt{2\pi t^3}}\, dt$$

$U(dt)$

$\parallel$

$E[e^{-tU}]$

Fréchet (ID).

Boschasimo

$$\frac{1}{\Gamma(a+s)} = \exp \int_0^\infty (e^{-sx} + \lambda x - 1)\, \frac{e^{-ax}}{x}\, dx \qquad \text{Lévy}$$

$$\downarrow$$

$$= \exp \int_0^\infty (1 - e^{-\frac{1}{2}s^2 t})\, \nu(dt)$$

$$\parallel$$

$$= \exp \int_0^\infty (1 - e^{-\frac{1}{2}s^2 t})\, E[e^{-tU}]$$

$$\downarrow$$

$$\text{"}H\text{"}(y\, g(u))$$

$g^{-1}(u) = -$

$$\parallel$$

$$e$$

(sin –)

$$\frac{1-e^{-\frac{1}{2}x^2/t}}{\sqrt{t}} = \int_0^\infty e^{-tz}\, \frac{2\sin^2(x\sqrt{z/2})}{\sqrt{\pi z}}\, dz$$

$g^{-1}(z)$.

---

x ~

$g(-1)$.

$$\frac{t^s}{\Gamma(a+1)} = \int e^{-s^2} __ .$$

$+ __ .$ DP → Brownian

⋮